DAY 2:

STRUCTURE OF AI SYSTEMS

The **structure of an Artificial Intelligence (AI) system** describes how an AI system receives information from the environment, processes it, makes decisions, and produces an action or output. It is best visualized as a five-layer stack or a layered pyramid where raw data transforms into intelligent action.

1. Infrastructure Layer :

- **Function:** Provides physical hardware, computing power, storage, and networking.
- **Components:** GPUs, TPUs, CPUs, cloud platforms (AWS, Azure, GCP), and edge devices.

2. Data Layer

- **Function:** Collects, cleans, governs, and stores the raw information needed for training and execution.
- **Components:** Enterprise data warehouses, data lakes, and vector databases.

3. Model Layer

- **Function:** Acts as the intelligence core where machine learning, deep learning, or foundational models (like transformers) identify patterns.
- **Components:** Open-source or proprietary models (e.g., GPT, Claude, Llama).

4. Context, Retrieval, and Orchestration Layer

- **Function:** Connects models to external data sources, manages workflows, and handles Retrieval-Augmented Generation (RAG).

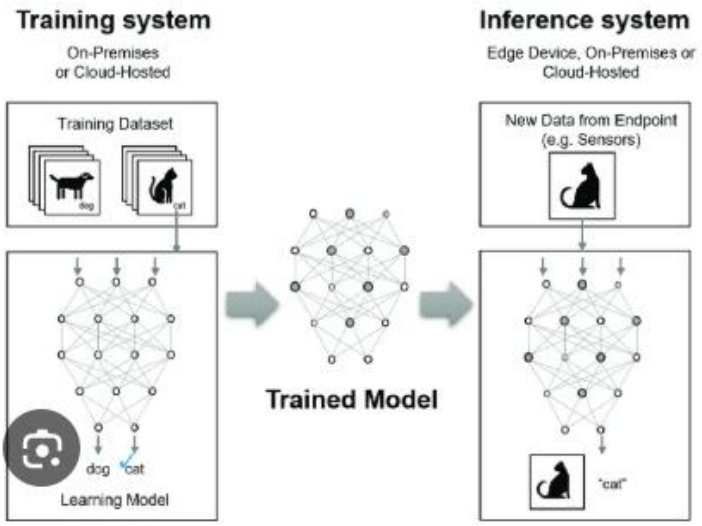
  (RAG is a technique that improves AI answers by letting a large language model search an external database or document library for real facts before it writes a response.)
- **Components:** APIs, data pipelines, agentic loops, and memory management.

5. Application and Presentation Layer

- **Function:** Delivers the functionality to end-users through interfaces and business logic.
- **Components:** Chatbots, recommendation engines, and user dashboards.

6. Governance and Operations (Ops) Layer

- **Function:** Monitors performance, evaluates outputs, and ensures security, compliance, and ethical standards.
- **Components:** LLM Ops tools, bias detection, and access control policies.



Real World Implications of AI

Artificial intelligence (AI) is used in everyday life and across industries to automate tasks, analyse large amounts of data, and make fast decisions.

Healthcare and Diagnostics

- **Medical Imaging:** AI algorithms scan X-rays, MRIs, and CT scans to help doctors find anomalies, tiny fractures, or early signs of cancer.
- **Drug Discovery:** Pharmaceutical companies use AI to review chemical compounds and identify promising drug candidates much faster than traditional methods.
- **Automated Disinfection**: Systems coordinate autonomous robots equipped with UV-C (ultraviolet-C light) to clean and kill germs and infection in hospital spaces instantly.

Transportation and Navigation

- **Traffic Routing:** Apps like Google Maps analyze live vehicle data to estimate travel times and suggest faster routes.
- **Autonomous Driving:** Self-driving systems use computer vision and sensors to read traffic signs, detect obstacles, and navigate roads safely.
- **Smart Cities:** Traffic control systems adjust signals dynamically based on real-time congestion and flow.

Finance and Banking

- **Fraud Detection**: Banks use AI to monitor transaction patterns and flag or decline suspicious activity immediately.

- **Algorithmic Trading:** Financial systems analyze market trends to execute trades in real time.
- **Customer Support:** Virtual assistants and banking chatbots help customers check account details and complete basic transactions.

Daily Consumer Tools

- **Recommendation Engines**: Streaming services and online stores use AI to suggest videos, music, or products based on user history.
- **Voice Assistants**: Tools like Siri and Alexa interpret spoken commands to answer questions, play music, or manage smart home settings.
- **Spam Filtering:** Email platforms use pattern recognition to keep unwanted messages out of your inbox.

AI Revolution and Current Trends In artificial Intelligence

The AI Revolution is the rapid transformation of technology, business and society driven by advances in machine learning, deep learning, generative AI, multimodal systems, AI agents, robotics and increasingly capable AI models. It is **accelerated** due to more data, more computing power (Cloud computing, GPU's, TPU's) and better algorithms (Deep learning, neural networks, transformers, Generative models)

Earlier AI systems were mainly designed for **specific tasks**.

Today, AI systems can work with **text, images, audio, video, code and other data**, and increasingly can perform multi-step tasks.

Current Trends in AI:

1. **Generative AI:** It **creates new content** rather than simply classifying existing data. It can generate Text, Images, Audio, Video, code, 3D content.
   Examples include AI chatbots, image generators and coding assistants.

2. **Multimodal AI:** Modern AI systems increasingly work with **multiple types of information simultaneously**. A system may receive an **image + text question** and generate a textual explanation.

3. **Agentic AI:** Agentic AI refers to artificial intelligence systems that operate autonomously to achieve goals with minimal human oversight. It follows the steps:

```
Understand Goal
      ↓
Plan Steps
      ↓
Use Tools
      ↓
Perform Actions
      ↓
Check Results
      ↓
Adjust Plan
      ↓
Complete Goal
```

Agentic AI takes an objective, makes a plan, uses outside tools, and fixes its own mistakes until the job is done.

4. **AI Reasoning:** It is the ability to perform **multi-step reasoning**. Instead of simply generating an immediate answer, models are increasingly designed to spend more computation on difficult tasks such as:
    - Mathematics
    - Programming
    - Scientific problems
    - Planning
    - Complex decision-making

This is helping AI move from: **Pattern recognition → reasoning and problem solving.**

5. **AI in Robotics:** AI is increasingly being combined with robotics.

Traditional Robot :

```
Fixed Instructions
       ↓
Fixed Action
```

AI- Powered Robot

```
Sensors
   ↓
Perception
   ↓
AI Model
   ↓
Reasoning / Planning
   ↓
Action
   ↓
Feedback
```

Applications are manufacturing, healthcare, agriculture, autonomous vehicles, service robots

Intelligent Agents:

An **Intelligent Agent** is like a smart assistant that **observes its environment**, **makes decisions**, and **takes actions** to achieve goals. Think of it as the brain inside a self-driving car, a chatbot, or even a game character — always sensing, thinking, and responding.

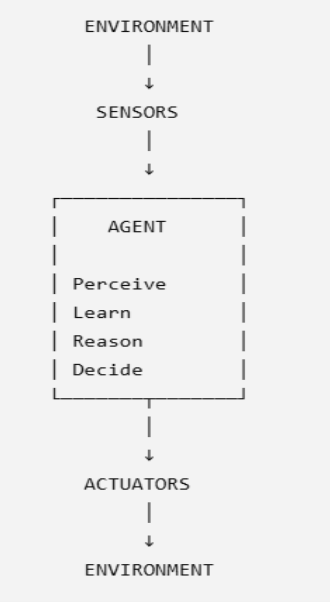
PERCEIVE → THINK/DECIDE → ACT

“What does your phone do when you say, ‘Set an alarm for 6 AM’?”

It: **hears you → understands the request → decides what to do → sets the alarm.**

That is the basic idea of an **agent**.

Structure of an Intelligent Agent



- **Sensors** → allow the agent to **perceive** the environment. Gather information (camera, microphone, data input).
- **Actuators** → allow the agent to **act** on the environment. Perform actions (move, speak, display results).

Agent	Sensors / Input	Decision	Action
Self-driving car	Cameras, radar, GPS	Brake / turn / accelerate	Controls vehicle
Robot vacuum	Camera, sensors	Where to clean	Moves & cleans
Chess AI	Board position	Which move is best?	Makes a move
Navigation system	Location, traffic	Which route is best?	Recommends route
Chatbot	User's text	What response is appropriate?	Generates response
Fraud detection system	Transaction data	Fraud or genuine?	Flags transaction

An intelligent agent should be able to **select appropriate actions based on what it knows and observes**

A simple robot:

If obstacle → stop.

An intelligent robot might:

Detect obstacle → identify it → evaluate alternatives → find another route → continue toward its destination.

Rational Agent

A **rational agent** chooses the action that is expected to produce the **best outcome**, based on the information available to it.

Example: Self-driving car

If the car sees Pedestrian crossing, Vehicle ahead, Red light, then It decides to

Slow down → Stop

Because that action best supports its goal of **safe travel**.

Rational ≠ perfect

(The agent may not know everything. It simply makes the **best decision possible with the information available.)**

PEAS Framework

A very useful framework for describing an intelligent agent is **PEAS**.

P → Performance Measure (How do we judge success?)

E → Environment (Where does the agent operate?)

A → Actuators (How does the agent act?)

S → Sensors (How does the agent perceive?)

Example: Self-driving car

P – Performance	Safety, travel time, comfort, fuel efficiency
E – Environment	Roads, traffic, pedestrians, weather
A – Actuators	Steering, brakes, accelerator
S – Sensors	Cameras, radar, GPS, other sensors

TYPES OF INTELLIGENT AGENTS

Type	Description	Example
Simple Reflex Agent	Acts only on current input	Thermostat adjusting temperature
Model-Based Agent	Uses internal model of the world	Self-driving car predicting traffic

Type	Description	Example
Goal-Based Agent	Chooses actions to reach goals	Chess AI planning moves
Utility-Based Agent	Maximizes happiness or efficiency	Recommendation system
Learning Agent	Improves with experience	Chatbot learning from conversations

Difference between the AI model and AI Agent:

An AI model may make a prediction, but an intelligent agent uses information to decide and potentially act toward a goal.

.

INTELLIGENT SYSTEMS

Early AI explored how intelligent behaviour could be produced using rules and expert knowledge. Later, researchers studied systems that could learn through interaction with their environment.

Artificial Intelligence is not limited to machines following predefined instructions. AI systems can demonstrate intelligent behavior in different ways—by learning from experience, interacting with their environment, following rules, using expert knowledge, or working collectively with other intelligent agents.

The important concepts here are:

- **Artificial Life** explores how life-like characteristics such as adaptation, evolution, and self-organization can be simulated in artificial systems.
- **Learning through Interaction** allows an AI system to improve its behaviour by interacting with the environment and learning from feedback.
- **Swarm Intelligence** shows how simple agents can work together and produce intelligent collective behaviour.
- **Emergent Behaviour** occurs when complex behaviour develops from simple rules and interactions among multiple agents.
- **Rule-Based Systems** make decisions using predefined **IF–THEN rules**.
- **Expert Systems** use expert knowledge and reasoning mechanisms to solve problems in a specific domain.

What is Artificial Life? (A Life)

Artificial Life is the field of AI and computer science that studies how **life-like behaviour can be created or simulated using computers, mathematical models, or robots**.

The goal is not necessarily to create real biological life, but to understand how **life-like behaviour can emerge from simple rules**.

Artificial Life (ALife) is the study of artificial systems that simulate or exhibit characteristics of natural living systems such as adaptation, evolution, reproduction, self-organization, and interaction with the environment.

Artificial Life = Creating or simulating life-like systems using artificial means.

Three Main Types of Artificial Life

- **Soft ALife:** Uses computer software and digital simulations (like genetic algorithms and virtual creatures) to model life inside a machine.
- **Hard ALife:** Builds physical robots and hardware that move, react, and adapt to the real world like living organisms.
- **Wet ALife:** Uses chemistry and synthetic biology to engineer artificial cells or biochemical systems that can grow and self-replicate.

How Artificial Intelligence powers Artificial Life

- **Generative Biology:** Advanced AI models can read and write DNA like a language, allowing scientists to generate synthetic proteins and design functional biological tools from scratch.
- **Evolutionary Computation:** Programs use rules of natural selection and mutation to let digital agents adapt to virtual environments over time.
- **Swarm Intelligence:** Swarm Intelligence is the study of how simple individuals, by interacting with each other and their environment, can collectively produce intelligent and efficient behaviour. Algorithms mimic social organisms, such as ant colonies or flocking birds, to coordinate groups of drones or virtual entities.

Artificial Life vs Artificial Intelligence

Artificial Intelligence	Artificial Life
Focuses on creating intelligent behavior	Focuses on creating/simulating life-like behavior
Usually concerned with tasks such as prediction, reasoning, planning	Studies adaptation, evolution, reproduction, self-organization
Example: Chatbot	Example: simulated artificial organisms
Machine learning is a major approach	Evolution and interaction are major themes

There is significant overlap between the two fields.

Learning Through Interaction:

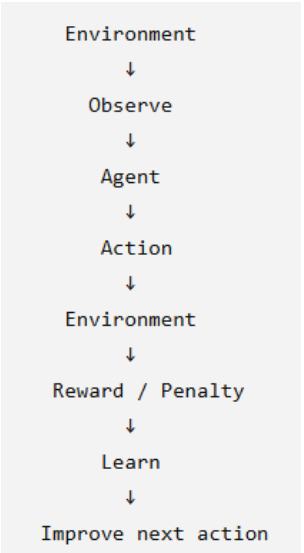
Learning through interaction means an AI system learns by **interacting with its environment**, taking actions, receiving feedback, and improving its future decisions.

This is the basic idea behind **Reinforcement Learning (RL)**.

For example, a chess-playing AI:

- Makes a move → gets feedback
- Winning move → positive reward
- Poor move → negative feedback
- Learns which actions lead to better results

Simple example: Game-playing AI, Robotics, Autonomous vehicle



Interaction + Feedback → Learning → Better decisions

Emergent Behavior

Emergent behavior occurs when **complex behaviour develops from simple rules and interactions among individual agents**.

No single agent needs to understand the complete behaviour.

Simple rules + Interaction → Complex collective behavior

Example:

Ant colony : Each ant follows simple rules. No ant is instructed. Yet the colony collectively discovers an efficient path.

```
Find food
   ↓
Leave pheromone
   ↓
Follow strong pheromone trails
   ↓
Other ants follow
   ↓
Efficient path emerges
```

Rule-Based Systems

A **rule-based system** makes decisions using predefined **IF–THEN rules**. This system is simple, predictable, easy to understand, easy to explain.

Rules are explicitly written by humans.

Example1:

```
IF
   fever = TRUE
   AND cough = TRUE

THEN
   possible_infection = TRUE
```

Example2:

```
IF
   fever = TRUE
   AND cough = TRUE

THEN
   possible_infection = TRUE
```

Basic structure

Input → Check conditions → IF-THEN Rules → Output

Expert Systems

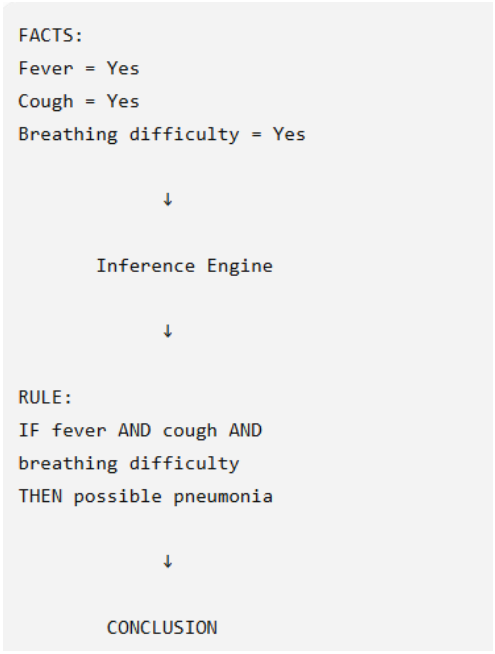
An **Expert System** is an AI system that uses **expert knowledge and reasoning to solve problems in a specific domain**.

It is essentially a more sophisticated application of knowledge-based/rule-based reasoning.

Knowledge Base contains Facts, Rules, Domain knowledge.

Inference Engine uses those facts and rules to reach a conclusion.

Example: Medical diagnosis



Concept	Main Idea	Example
Learning through Interaction	Learn from actions and feedback	Game-playing AI
Emergent Behavior	Complex behavior emerges from simple interactions	Ant colony
Rule-Based System	Uses predefined IF–THEN rules	Temperature control
Expert System	Uses expert knowledge + reasoning to solve a domain problem	Medical decision-support system